

# Gradient Descent. Convergence rates

## Seminar

Optimization for ML. Faculty of Computer Science. HSE University

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The bottleneck (for almost all gradient methods) is choosing step-size, which can lead to the dramatic difference in method's behavior.

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- **Backtracking line search.** Fix two parameters:  $0 < \beta < 1$  and  $0 < \alpha \leq 0.5$ . At each iteration, start with  $t = 1$ , and while

$$f(x_k - t\nabla f(x_k)) > f(x_k) - \alpha t \|\nabla f(x_k)\|_2^2,$$

shrink  $t = \beta t$ . Else perform Gradient Descent update  $x_{k+1} = x_k - t\nabla f(x_k)$ .

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- **Exact line search.**

$$\eta_k = \arg \min_{\eta \geq 0} f(x_k - \eta \nabla f(x_k))$$

## Direction of local steepest descent

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The result of this method is

$$x_{k+1} = x_k - \alpha f'(x_k)$$

## Minimizer of Lipschitz parabola

If a function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is continuously differentiable and its gradient satisfies Lipschitz conditions with constant  $L$ , then  $\forall x, y \in \mathbb{R}^n$ :

$$|f(y) - f(x) - \langle \nabla f(x), y - x \rangle| \leq \frac{L}{2} \|y - x\|^2,$$

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which geometrically means, that if we'll fix some point  $x_0 \in \mathbb{R}^n$  and define two parabolas:

$$\phi_1(x) = f(x_0) + \langle \nabla f(x_0), x - x_0 \rangle - \frac{L}{2} \|x - x_0\|^2,$$

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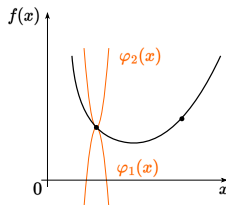


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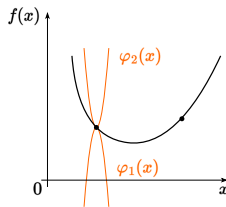


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$$\nabla \phi_2(x) = 0$$

$$\nabla f(x_0) + L(x^* - x_0) = 0$$

$$x^* = x_0 - \frac{1}{L} \nabla f(x_0)$$

$$x_{k+1} = x_k - \frac{1}{L} \nabla f(x_k)$$

This way leads to the  $\frac{1}{L}$  stepsize choosing. However, often the  $L$  constant is not known.

# Any $\mu$ -strongly convex differentiable function is a PL-function

## Theorem

If a function  $f(x)$  is differentiable and  $\mu$ -strongly convex, then it is a PL function.

## Proof

By first order strong convexity criterion:

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{\mu}{2} \|y - x\|_2^2$$

Putting  $y = x^*$ :

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Let  $a = \frac{1}{\sqrt{\mu}} \nabla f(x)$  and

$$b = \sqrt{\mu} (x - x^*) - \frac{1}{\sqrt{\mu}} \nabla f(x)$$

Then  $a + b = \sqrt{\mu} (x - x^*)$  and

$$a - b = \frac{2}{\sqrt{\mu}} \nabla f(x) - \sqrt{\mu} (x - x^*)$$

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$$f(x) - f(x^*) \leq \frac{1}{2\mu} \|\nabla f(x)\|_2^2,$$

which is exactly the PL condition. It means, that we already have linear convergence proof for any strongly convex function.

## Exact line search aka steepest descent

$$\alpha_k = \arg \min_{\alpha \in \mathbb{R}^+} f(x_{k+1}) = \arg \min_{\alpha \in \mathbb{R}^+} f(x_k - \alpha \nabla f(x_k))$$

More theoretical than practical approach. It also allows you to analyze the convergence, but often exact line search can be difficult if the function calculation takes too long or costs a lot. Interesting theoretical property of this method is that each following iteration is orthogonal to the previous one:

$$\alpha_k = \arg \min_{\alpha \in \mathbb{R}^+} f(x_k - \alpha \nabla f(x_k))$$

Optimality conditions:

$$\nabla f(x_{k+1})^\top \nabla f(x_k) = 0$$

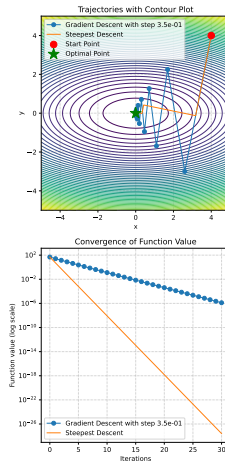


Figure 2: Steepest Descent

Open In Colab 



## Convergence analysis. Backtracking line search

Assume that  $f$  is convex, differentiable and Lipschitz gradient with constant  $L > 0$ .

### Theorem

Gradient descent with fixed step size  $t \leq 1/L$  satisfies

$$f(x^{(k)}) - f^* \leq \frac{\|x^{(0)} - x^*\|_2^2}{2tk}$$

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Let  $y = x^+ = x - t\nabla f(x)$ , then:

$$f(x^+) \leq f(x) - \left(1 - \frac{Lt}{2}\right) t \|\nabla f(x)\|_2^2 \leq f(x) - \frac{1}{2L} \|\nabla f(x)\|_2^2$$

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This recalls us the stopping condition in Backtracking line search when  $\alpha = 0.5, t = \frac{1}{L}$ . Hence, Backtracking line search with  $\alpha = 0.5$  plus condition of Lipschitz gradient will guarantee us the convergence rate of  $O(1/k)$ .

# Problem

Consider the problem

$$\min_{x \in \mathbb{R}^n} f(x),$$

where  $f(x)$  is convex and  $L$ -smooth. Find convergence rate of gradient descent with optimal theoretical step size  $\eta_k = \frac{1}{L}$  for the *mean point* and for the *best point*. In other words get upper bounds on

- $f(\bar{x}_N) - f^*$ , where  $\bar{x}_N = \frac{1}{N} \sum_{i=0}^{N-1} x_i$ ,
- $\min_{0 \leq i \leq N-1} f(x_i) - f^*$ .

## i Gradient descent step

$$x_{k+1} = \arg \min_{x \in \mathbb{R}^n} \left\{ \Psi_k(x) \equiv f(x_k) + \langle \nabla f(x_k), x - x_k \rangle + \frac{L}{2} \|x - x_k\|_2^2 \right\}$$

# Problem

Consider the problem

$$\min_{x \in \mathbb{R}^n} f(x),$$

where  $f(x)$  is convex and  $L$ -smooth. Find convergence rate of gradient descent with optimal theoretical step size  $\eta_k = \frac{1}{L}$  for the *mean point* and for the *best point*. In other words get upper bounds on

- $f(\bar{x}_N) - f^*$ , where  $\bar{x}_N = \frac{1}{N} \sum_{i=0}^{N-1} x_i$ ,
- $\min_{0 \leq i \leq N-1} f(x_i) - f^*$ .

## i Gradient descent step

$$x_{k+1} = \arg \min_{x \in \mathbb{R}^n} \left\{ \Psi_k(x) \equiv f(x_k) + \langle \nabla f(x_k), x - x_k \rangle + \frac{L}{2} \|x - x_k\|_2^2 \right\}$$



Tip

Use the fact that  $\Psi_k(x)$  is  $L$ -strongly convex due to quadratic regularizer.

# Code

Examples:  code snippet.